

4.1

(a.)(b.)(c.)

$$\begin{aligned}
 a. \quad R^2 &= \frac{SSR}{SST} = \frac{SSR}{\sum (y_i - \bar{y})^2} = \frac{SSR}{\sum y_i^2 - N\bar{y}^2} \\
 &= \frac{375.47}{1825.94 - 20 \times (19.21)^2} \\
 &= 0.8429
 \end{aligned}$$

$$b. \quad R^2 = 0.7911 = \frac{SSR}{\underset{\substack{\uparrow \\ SST}}{1825.94}} \quad SSR = 574.29$$

$$SST = SSR + SSE$$

$$\begin{aligned}
 SSE &= 1825.94 - 574.29 \\
 &= 1251.65
 \end{aligned}$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{1251.65}{20-2} = 8.425$$

$$c. \quad \underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum e_i^2}_{SSE}$$

$$631.63 = SSR + 182.85$$

$$SSR = 448.78$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63}$$

$$= 0.7105$$

4.3

(a.)(b.)(c.)(d.)(e.)

a.

$$\hat{y}_0 = 1.2 + 0.8x_0$$

$$= 1.2 + 0.8 \times 4$$

$$= 4.4$$

b.

$$\bar{x} = (3+2+1+(1)+0) : 5 = 1$$

$$se(f) = \sqrt{\widehat{Var}(f)} = \sqrt{s^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]}$$

$$= \sqrt{\frac{se_{\hat{x}}^2}{N-2} \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]}$$

$$= \sqrt{\frac{3.6}{5-2} \left(1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right)}$$

$$= \sqrt{1.2 \times \frac{21}{10}}$$

$$= \sqrt{2.52}$$

$$= 1.5875$$

c. 查t值表 $t_{(n-1), \alpha/2}$

$$t_{3, 0.025} = 3.182$$

$$[\hat{y}_0 - t_{3, 0.025} \times se(f), \hat{y}_0 + t_{3, 0.025} \times se(f)]$$

$$= [4.4 - 3.182 \times 1.5875, 4.4 + 3.182 \times 1.5875]$$

$$= [-0.6514, 9.4514]$$

d.

$$t_{3, 0.005} = 5.841$$

$$[4.4 - 5.841 \times 1.5875, 4.4 + 5.841 \times 1.5875]$$

$$= [-4.8726, 13.6726]$$

e.

$$s_e(f) = \sqrt{1.2 \times \left(1 + \frac{1}{5} + \frac{(1-1)^2}{10}\right)}$$

$$= \sqrt{1.44}$$

$$= 1.2$$

$$\bar{y} = (4 + 2 + 3 + 1 + 0) / 5 = 2$$

$$\left[2 - 3.182 \times 1.2, 2 + 3.182 \times 1.2 \right]$$

$$= [-1.8184, 5.8184]$$

c. interval width = 10.10285

d. interval width = 7.6368 [✓]

以題區間寬度較寬，

因當 x 離 $\bar{x}$ 愈遠，預測不確定性[se(f)]

愈大。

4.25

(a.)

Call:

```
lm(formula = log(price) ~ sqft, data = collegetown)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.32528	-0.19199	0.00122	0.21678	0.86609

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.393866	0.043288	101.50	<2e-16 ***
sqft	0.036044	0.001486	24.26	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom

Multiple R-squared: 0.5417, Adjusted R-squared: 0.5408

F-statistic: 588.7 on 1 and 498 DF, p-value: < 2.2e-16

(a)

slope at the sample mean is : 9.019661

elasticity at the sample mean is : 0.9833701

b1: 當 sqft=0 , price 估計值為 $\exp^{4.393866}=80.9528$

b2: 當 sqft 增加 1 單位 (以百為單位), price 估計增加 3.6%

(b.)

Call:

```
lm(formula = log(price) ~ log(sqft), data = collegetown)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.36864	-0.20849	0.00487	0.23204	1.21099

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.04965	0.15797	12.97	<2e-16 ***
log(sqft)	1.02483	0.04839	21.18	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3643 on 498 degrees of freedom

Multiple R-squared: 0.4738, Adjusted R-squared: 0.4728

F-statistic: 448.5 on 1 and 498 DF, p-value: < 2.2e-16

(b)

slope at the sample mean is : 9.399923

elasticity at the sample mean is : 1.024828

a1: 當 $\text{sqft}=1$ [$\ln(\text{sqft})=0$] , price 估計值為 $\exp^{2.04965}=7.7652$

a2: 當 sqft 增加 1% , price 估計增加 1.02483%

(c.)

(c)

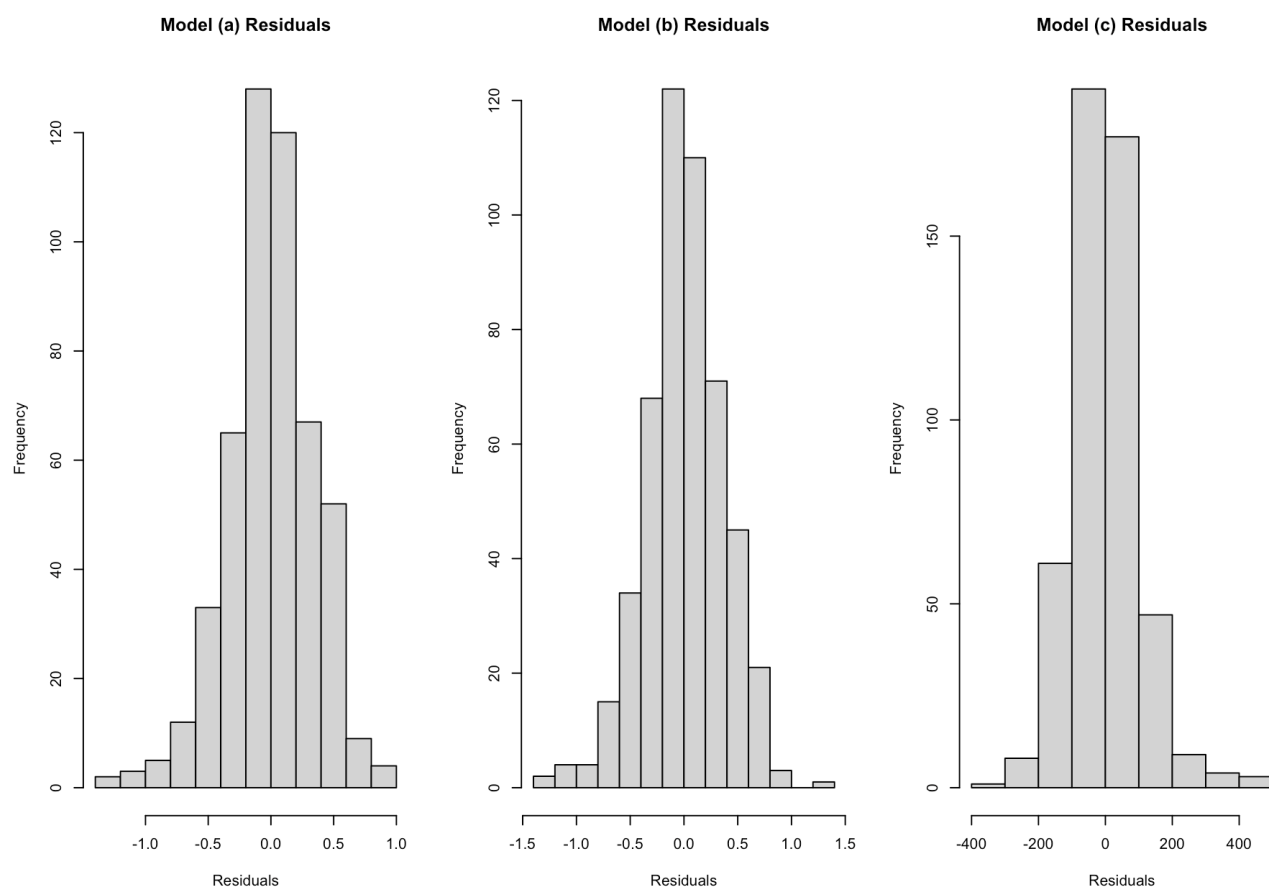
R-squared value from the linear model is : 0.6413167

Generalized R-squared from (b) is : 0.6445084

Generalized R-squared from (c)(the linear model) is : 0.6413167

Model(b)[log-log model]的 R-squared 略大，應較佳。

(d.)



(d)

the Jarque-Bera statistics for (a) is : 26.67899

the Jarque-Bera statistics for (b) is : 14.27871

the Jarque-Bera statistics for (c) is : 221.1115

(d)

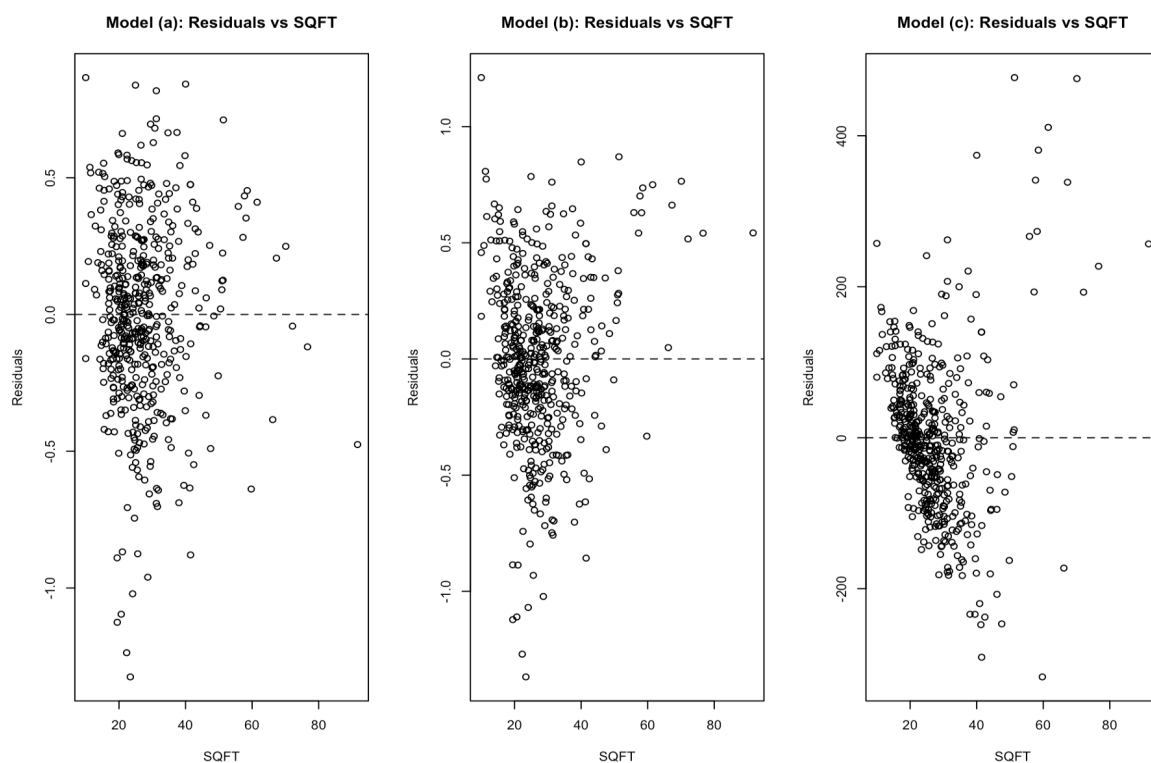
p-value for (a) is : 1.609651e-06

p-value for (b) is : 0.0007932632

p-value for (c) is : 0

3 個模型的 p 值皆很小，故殘差分佈不全然服從常態分佈。而三者之中又以 model(b)的 p 值相對較大且 JB 檢定值較小，應相對接近常態分佈。

(e.)



Model(c)在 SQFT 值較大時，殘差變動幅度有增加趨勢。Model(b)對自應變數取 log 後，應為三者之中，殘差最為呈現隨機分佈，且在不同 SQFT 值的變動幅度較無太大差異。

(f.)

(f)

predict value for model (a) is : 214233.6

predict value for model (b) is : 227538.6

predict value for model (c) is : 246455.7

(g.)

(g): 95%Prediction Interval

model (a) log-linear: 109.7685,418.1167

model (b) log-log: 111.1406,465.8406

model (c) linear : 44.27727,448.6341

(h.)

Linear model 的 JB 檢定值太高，故不適合。而另兩模型在 R-squared 差不多的情形下，Model(b)[log-log model]的 JB 檢定值更小且 p 值更大，故較適合。